



Guideline

Cardiac Monitoring

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Nursing and Medical staff

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Introduction

Cardiac monitoring allows for risk stratification of varying patient presentations including conduction abnormalities and cardiac dysrhythmias, acute coronary syndromes (ACS), severe metabolic derangements, toxicity, syncope, structural cardiac disease, and patients at risk of sudden cardiac death. Cardiac monitoring is also used to assess the effectiveness and/or side effects of medication. The causes of electrical abnormalities are broad, and areas other than those sitting under cardiac or critical care services may need to utilise cardiac monitoring. Clinicians supervising the patient must be skilled in rhythm recognition. These guidelines are aimed primarily at medical and nursing healthcare professionals who are working in areas where cardiac monitoring is available.

The purpose of this Guideline is to provide the minimum recommended standards for initiation and removal of cardiac monitoring. The document should be used in conjunction with area specific guidelines and policies. Due to the variety of patient cohorts and clinical areas where cardiac monitoring may be undertaken, this document does not stipulate absolute criteria for when to initiate and remove cardiac monitoring and in no way replaces the need for clinical assessment of individual patients.

Clinicians trained in rhythm recognition can care for patients requiring cardiac monitoring in areas including Operating Theatres, Invasive Procedural Units (eg; CVIL), Coronary Care Unit (CCU), Cardiology ward (G41), Emergency Department (ED), Cardio-Thoracic Ward (G62), Intensive Care Unit (ICU) and the Medical Assessment Unit (MAU). Cardiac monitoring can be performed in other locations provided there are staff trained in rhythm recognition, such as in a Medical Emergency via a defibrillator or portable monitor.

Cardiac monitoring should be performed continuously via direct visualisation (ie; bedside monitor). In certain areas such as MAU, and in appropriate patient cohorts, cardiac monitoring may be performed via indirect visualisation (ie; telemetry units with a central monitoring station). In these instances, telemetry monitoring alarms are reviewed by medical staff retrospectively once per day. Ward patients who require real time monitoring should be monitored in an appropriate critical/acute care area (as above) with staff trained in rhythm recognition.

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☐	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☐	Impact on Patient Safety	☒
Breach professional standards	☐	Misconduct	☐
Breach SCGH Mission & Values	☐	Staff Safety	☐
Other:	☐		

1.0 Cardiac Monitoring Requirements

1.1 Staff training requirements

All patients who require cardiac monitoring via direct visualisation in critical care areas will be cared for by a team of clinicians where at least 75% of the nursing staff on shift have a current Hospital Advanced Life Support (HALS) competency unless otherwise agreed (ie; G62 maintain HALS competency for 33% of their nursing staff to ensure their 4 bed high dependency unit is staffed at all times by 4 HALS competent nursing staff).

1.2 Initiation of cardiac monitoring

Registered nurses may initiate cardiac monitoring in the absence of medical direction, though the treating medical officer is responsible for determining the ongoing requirement for cardiac monitoring.

1.3 Alarms

All alarms need to be responded to by the closest nurse / doctor who must:

- Assess patient using an ABCDE approach.
- Perform 12 lead ECG as soon as practicable.
- Escalate according to the SCGOPHCG Hospital Policy # HP234 - Recognising and Responding to Acute Deterioration.
- If the patient is in cardiac arrest, follow the SCGOPHCG Guideline # CPG010 - In-Hospital Resuscitation.

1.4 Reviewing alarms

On commencing cardiac monitoring and every shift thereafter individualise alarm limits to reflect the patient's condition/diagnosis, and change as required:

- At a minimum, review alarm limits at the start of each shift to prevent alarm fatigue.
- Review the patient alarm history each shift and delete any inappropriate alarms.
- If the patient has a chronic arrhythmia adjust the alarm for that arrhythmia.
- If a patient has a pacemaker ensure pacing detection function is activated.

1.5 Documentation

- The cardiac rhythm must be documented at the commencement of cardiac monitoring and every shift thereafter.

- Any changes in the cardiac rhythm must be documented in the medical records with a 12 lead ECG attached.
- Cardiac (ECG) rhythm analysis must be performed with each set of observations and recorded on the Observation and Response Chart below the heart rate e.g. NSR, AF, and SVT etc. See glossary of terms for approved ECG rhythm abbreviations.

1.6 Ongoing cardiac monitoring

When cardiac monitoring has been ordered the treating team must review the need for ongoing monitoring daily and document this requirement in the medical records.

1.7 De-monitoring patients

Cardiac monitoring can only be ceased under the direction of a medical officer, who must document de-monitoring instructions in the medical records.

1.8 Escorting cardiac monitored patients

- Patients requiring cardiac monitoring via direct visualisation who need to be transported out of the designated cardiac monitoring area must have continuous monitoring via portable monitor/defibrillator.
- Patients who are cardiac monitored may only be escorted by appropriately trained staff that are skilled in rhythm recognition and the use of the equipment.
- Only HALS competent nurses or healthcare staff with Advanced Life Support (ALS) Level 2 certification may escort patients with cardiac monitoring.
- As a minimum ensure escort staff have access to a portable monitor and defibrillator, with consideration of bag valve mask, portable oxygen and suction depending upon patient acuity.

1.8.1 Dialysis Unit cardiac monitoring

- Patients requiring admission to the Dialysis Unit for an acute renal presentation such as an elevated serum potassium level with ECG changes require cardiac monitoring (please see below table 2.0 – Indications for Cardiac Monitoring).
- In circumstances where it is anticipated that dialysis will correct underlying monitoring requirements, the minimum duration for cardiac monitoring is until the acute electrolyte disturbance is corrected and there are no related arrhythmias present.
- Patients who are attending dialysis and are being cardiac monitored and/or admitted for reasons other than an acute renal presentation (ie; acute coronary syndrome) are required to be cared for by critical care nursing staff with HALS competency.

2.0 Indications for Cardiac Monitoring

Category I: Cardiac monitoring is strongly indicated		
INDICATIONS FOR MONITORING	MONITORING DURATION	COMMENTS
Acute Coronary Syndromes (ACS)		
ST Elevation Myocardial Infarction (STEMI) 'Not low risk' or 'high risk' Non-ST Elevation Myocardial Infarction (NSTEMI)*	Monitor for at least 24-48 hours. ^{1, 2}	Discussion with Cardiology. Consider ST segment monitoring if indicated. ¹

*Refer to High Risk Features for ACS in the SCGH Acute Coronary Syndrome (ACS) Assessment and Treatment Algorithm		
Multiple cardiovascular risk factors* with negative troponin *Refer to High Risk Features for ACS in the SCGH Acute Coronary Syndrome (ACS) Assessment and Treatment Algorithm	Monitor for at least 24-48 hours.¹	Discussion with Cardiology. Early risk stratification for future cardiac events with stress testing is advisable.⁷
Life Threatening Arrhythmias		
Tachyarrhythmias - Wide complex tachyarrhythmia including Ventricular Tachycardia (VT), Ventricular Fibrillation (VF) or supraventricular tachycardia (SVT) with aberrancy - Atrial Fibrillation (AF) or SVT with high risk of 1:1 conduction in presence of accessory pathway - Tachy-Brady Syndrome - Any tachyarrhythmia causing haemodynamic compromise[†]	Continuous monitoring until reversible causes are rectified, stabilisation of rhythm, and/or implantable cardioverter defibrillator (ICD) implantation.^{1, 3}	[†]Systolic BP below 90mmHg, symptoms of circulatory failure/cardiogenic shock, chest pain or syncope.
Symptomatic Bradyarrhythmias - Sinus node dysfunction with asystole of ≥3 seconds or evidence of chronotropic incompetence 2nd or 3rd degree Atrioventricular (AV) block - Syncope with evidence of conduction disease on ECG, including; - Trifascicular block - Bifascicular block (Left bundle branch block (LBBB) / Right bundle branch block (RBBB) with 1st degree heart block)	Monitor until reversible causes are identified and treated, symptoms have been stabilised, and/or a device is implanted and tested (minimum 24 hours).³	
Arrhythmia with known congenital or acquired channelopathy - Prolonged QTc (>500msec) with non-sustained ventricular tachycardia or torsades de pointes - Catecholaminergic Polymorphic Ventricular Tachycardia - Long QT Syndrome (LQTS)	Continuous monitoring until stabilisation of rhythm, appropriate medical therapy or reversible causes are rectified.^{1, 3}	

• Brugada Syndrome • Wolff-Parkinson-White (WPW) Syndrome		
Syncope with high risk clinical features	Continuous monitoring for at least 24 hours until cause is identified and treatment provided. ¹	
Post procedural monitoring e.g. vascular surgery, electrophysiology procedures, permanent pacemaker (PPM) or ICD implantation, major cardiac interventions, open heart surgery, etc.	Duration of monitoring varies based upon procedure and patient factors; monitor as per specific post procedure protocol or as per team written plan. ¹	
Miscellaneous Indications		
Post cardiac arrest	Continuous monitoring for a minimum of 24 hours and until the cause has been identified and treated. ³	
Medication administration, including: • Anti-arrhythmics • Pro-arrhythmic drugs • Inotropes • Vasoactive drugs • Fibrinolytics • Intravenous potassium at a rate of >10mmol per hour[†]	Continuous monitoring during infusion and loading. ^{1, 4} Prolonged monitoring if any evidence of ongoing unstable arrhythmia. ¹	Early cardiology referral if there is evidence of ongoing unstable arrhythmia. Refer to High Risk Medications Policy [†] Refer to Potassium Medication Management Guideline
Overdose of any medication leading to ECG changes with or without life threatening arrhythmias	Monitor until clinically stable and free of the influence of the drug(s) ¹ , or as per advice from Cardiology/Toxicology/inpatient team.	
Pericardial effusion with haemodynamic compromise ⁷	Continuous monitoring pre and post treatment as per cardiology/inpatient team review.	Urgent advice from Cardiology/inpatient team regarding pericardiocentesis.
Infective endocarditis with valve dysfunction or abscess	Until clinically stable ¹ , or as per advice from Cardiology/inpatient team.	
Blunt cardiac injury such as cardiac contusion	As per advice from Trauma/Cardiology team. ⁶	
Acute and severe electrolyte disturbances ^{5, 6} :	Monitor until the acute electrolyte disturbance is corrected and there are no related arrhythmias present. ¹	

- Serum potassium <3 mmol/L or >6 mmol/L with ECG changes - Serum magnesium <0.6 mmol/L or >2 mmol/L - Serum calcium (corrected) <1.8 mmol/L or >3 mmol/L		
Electrical injury	Monitor as per ED physician order. ⁶	

Category II: Cardiac monitoring is advised		
INDICATIONS FOR MONITORING	MONITORING DURATION	COMMENTS
Non-life threatening arrhythmias with haemodynamic compromise		
- New onset SVT or AF with rapid ventricular rate (>140bpm) - Chronic AF with increasing ventricular rate	Monitor until Cardiology/inpatient team review. ¹	
Non-life threatening arrhythmias without haemodynamic compromise		
Asymptomatic bradycardias with heart block	Monitor until Cardiology/inpatient team review. ¹	

Category III: Cardiac monitoring generally not indicated
- Terminally ill patients where arrhythmias will not be acted upon and comfort care is the treatment goal¹ - Patients with Goals of Patient Care for comfort treatment deemed unsuitable for any intervention¹ - Low risk NSTEMI¹ (refer to High Risk Features for ACS in the SCGH Acute Coronary Syndrome (ACS) Assessment and Treatment Algorithm) - Low risk, troponin negative chest pain¹ - Uncomplicated syncope with no high risk features or haemodynamic compromise, and unremarkable baseline ECG¹ - Chronic, haemodynamically stable atrial fibrillation¹ - Chronic premature ventricular beats in the haemodynamically stable patient¹ - Asymptomatic Wenckebach or transient AV block of vagal origin¹ - Any size pericardial effusion without haemodynamic compromise⁷ - Patients with a stable, functioning ICD/PPM who have had a post implant check¹ - Pre-existing rhythm devices (ICD/PPM) where admission is for an unrelated indication¹

Glossary of Terms

NSR – Normal Sinus Rhythm
 ST – Sinus Tachycardia
 SB – Sinus Bradycardia
 AF – Atrial Fibrillation
 AFI – Atrial Flutter
 JB – Junctional Bradycardia
 SVT – Supra-Ventricular Tachycardia
 AVNRT – Atrio-Ventricular Nodal Re-entry Tachycardia
 AVRT – Atrio-Ventricular Re-entry Tachycardia
 WPW – Wolff-Parkinson-White Syndrome
 VT – Ventricular Tachycardia
 pVT – Pulseless Ventricular Tachycardia
 VF – Ventricular Fibrillation
 CHB – Complete Heart Block
 1^o HB – First degree Atrio-Ventricular Block
 2^o HB # 2 – Second degree Atrio-Ventricular Block type 2 / Mobitz II
 2^o HB # 1 – Second degree Atrio-Ventricular Block type 1 / Wenkebach
 (A)IVR – (Accelerated) Idio-Ventricular Rhythm
 VPB – Ventricular Premature Beat
 APB – Atrial Premature Beat
 LBBB – Left Bundle Branch Block
 RBBB – Right Bundle Branch Block
 LAHB – Left Anterior Hemi Block
 LPHB – Left Posterior Hemi Block
 NSTEMI – Non ST Elevation Myocardial Infarction
 STEMI – ST Elevation Myocardial Infarction

Related Documents

- [SCGOPHCG In-Hospital Resuscitation Guideline](#)
- [SCGOPHCG Recognising and Responding to Acute Deterioration Policy](#)
- [Principles of ECG and Cardiac Monitoring Self Directed Learning Package](#)
- [Australian Commission on Safety and Quality in Health Care. Clinical Governance Standard](#)
- [SCGOPHCG High Risk Medications Policy](#)
- [SCGOPHCG Medication Management Guideline – Potassium Supplements - Intravenous](#)
- G72 Telemetry Area Specific Guideline

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